



Aprendizagem 2025
Lab 7-9: Neural Networks

Practical exercises

I. Multi-Layer Perceptron

1. Consider a network with three layers: 5 inputs, 3 hidden units and 2 outputs where all units use a sigmoid activation function.
 - a) Initialize connection weights to 0.1 and biases to 0. Using the squared error loss do a stochastic gradient descent update (with learning rate $\eta=1$) for the training example

$$\left\{x = [11000]^T, z = [10]^T\right\}$$

- b) Compute the MLP class for the query point $x_{new} = [10001]^T$
2. Consider a network with four layers with the following numbers of units 4, 4, 3, 3.

Assume all units use the *hyperbolic tangent* activation function.

- a) Initialize all connection weights and biases to 0.1. Using the squared error loss do a *stochastic gradient descent* update (with learning rate $\eta=0.1$) for the training example:

$$\left\{x = [1010]^T, z = [010]^T\right\}$$

- b) Reusing the computations from the previous exercise do a *gradient descent update* (with learning rate $\eta=0.1$) for the batch with the training example from the a) and the following:

$$\left\{x = [00100]^T, z = [001]^T\right\}$$

- c) Consider the learned MLPs from a) and b). Which has smallest squared error? Which model has better classification accuracy?
- d) Compute the MLP class for the query point $x_{new} = [1110]^T$

3. Repeat the exact same exercise, but this time with following adaptations:

- the output units have a *softmax* activation function
- the error function is *cross-entropy*

What are the major differences between using squared error and cross-entropy?

II. Model Complexity (optional)

4. [optional] For the following scenarios which has the smallest number of parameters?

a) three-dimensional real inputs classified by

- MLP with one hidden layer with the following units per layer 3 2 2
- simple Bayesian classifier with multivariate gaussian likelihood function

b) N -dimensional real inputs classified by

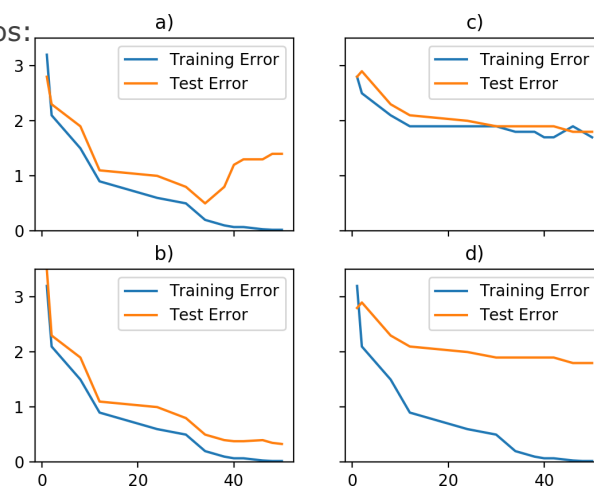
- perceptron
- MLP with two hidden layers with the following units per layer N , $\frac{N}{2}$, $\frac{N}{2}$, 2
- naive Bayes with Gaussian likelihoods
- simple Bayesian classifier with multivariate gaussian likelihood function

5. [optional] Choose between increase, decrease, maintain for each of the following factors:

- training data
- regularization
- number of parameters

For each of the following four scenarios:

Justify each decision.



Programming quest

Resources: https://scikit-learn.org/stable/modules/neural_networks_supervised.html
as well as *Classification*, *Regression* and *Evaluation* notebooks

6. Consider a 10-fold CV, and MLPs with a single hidden layer with 5 nodes. Using *sklearn*:
 - a) assess the classification accuracy of the MLP on the *iris* data using a cross-entropy loss
 - b) assess the MAE of the MLP on the *housing* data using a squared error loss